(vi) $L(x)$: Number of persons lived in aggregate in the age interval $(x,x+1)$

$$i.e.,\ L(x) = \frac{l(x)+l(x+1)}{2}$$

or equivalently

$$L(x) = l(x) - \frac{1}{2}d(x)$$

(vii) $T(x)$: Number of years lived by the ***cohort*** at and above the age x

$i.e.,\ T(x) = L(x)+L(x+1)+L(x+2)+ \ldots$

or equivalently

$T(x+1) = T(x) - L(x)$.

Total number of years lived by the ***cohort*** after x years of age.

(viii) $e^0(x)$: Expectation of life

$$e^0(x) = \frac{T(x)}{l(x)}$$

It gives the average number of years a person of age x years is expected to survive under the existing mortality conditions.

Assumptions of Life Table

The following assumptions are made while constructing a Life Table.

(i) There is no effect of immigration and emigration on the ***cohort***. It means that the reduction in the number of the initial ***cohorts*** is merely due to deaths.

(ii) The size of ***cohort*** begins with a convenient figure, it is conventionally 1,00,000.

(iii) Deaths are uniformly distributed over each age interval.

Example 8.6

A Life Table was constructed for a ***cohort***. The following is a section of the table, wherein some of the entries are not available. Find the estimates of missing values and complete the Life Table.

Age (in years)	$l(x)$	$d(x)$	$p(x)$	$q(x)$	$L(x)$	$T(x)$	$e^0(x)$
40	10, 645	-	-	-	-	1, 93, 820	-
41	10, 543	169	-	-	-	-	-

Solution:

The Life Table can be completed using the relationship among missing terms and other terms.

The number of persons who die before reaching age 40 years is calculated as

$$d(40) = l(40) - l(41)$$
$$= 10645 - 10543$$

Therefore, $d(40) = 102$.

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